LOCO-Inspect AI: Multi-Category Industrial Visual Anomaly Detection & Defect Localization System

Graduate Certificate in Pattern Recognition Systems — Technical Final Defense Presentation

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1. Mathematical Formulation & System Overview

Unsupervised Open-Set Formulation

- Problem Space: Input space X in R^{H x W x C}. Normality manifold M = {x in X | p_nom(x) > theta_0}.
- Training Distribution: Learn support supp(p_nom) purely from defect-free nominal images D_train without peeking at anomaly set A.
- Inference Objective: Compute distance metric s(x) = inf_{m in M} d(phi(x), phi(m)) to determine anomaly decision y in {0, 1}.
- Zero-Defect Escape Requirement: High-sensitivity calibration guaranteeing Recall >= 95% on mission-critical manufacturing lines.

System Pipeline Architecture

- Backbone: WideResNet-50-2 tapping intermediate multi-scale feature hierarchies (Layers 2 & 3).
- Feature Manifold: Concatenated patch representation generating 784 localized vectors of R^{1536} per image.
- Compression Engine: Johnson-Lindenstrauss Random Projection (128-D) + Greedy Minimax Coreset Subsampling (10%).
- Execution Speed: Chunked GPU TorchNN search running at ~85 ms per frame (11.8 FPS) on NVIDIA RTX 5070.

2. Structural vs. Logical Anomaly Formalisms

Structural Anomaly Formulation

- Physical Defect Model: Localized high-frequency perturbation x_tilde = x + delta where supp(delta) is compact.
- Manifestations: Scratches, cracks, punctures, contamination, and structural material deformities.
- Manifold Property: Perturbs local patch feature vector p_i outside the nominal patch manifold M_p.
- Detection Mechanism: Readily flagged by nearest-neighbor Euclidean distance in deep feature space.

Logical Anomaly Formulation

- Relational Rule Model: Represent product as a spatial graph G = (V, E) of components and spatial relations.
- Manifestations: Missing parts (e.g., label), quantity violations (screw counts), inverted orientations, wrong locations.
- Manifold Property: Individual patches p_i remain inside nominal patch manifold M_p; violation occurs in relational topology E.
- The Core Challenge: Pure permutation-invariant nearest-neighbor visual search is blind to global relational topology.

3. Deep Representation: Multi-Scale Patch Embedding

WideResNet-50-2 Hierarchical Feature Tapping

- Input Preprocessing: Standardization to 256x256x3, center-crop, ImageNet tensor normalization (mu, sigma).
- Layer 2 Activations: Spatial dimension 28x28 with 512 channels. Effective receptive field RF ~ 39x39 px.
- Layer 3 Activations: Spatial dimension 14x14 with 1,024 channels. Effective receptive field RF ~ 79x79 px.
- Dual-Scale Advantage: Layer 2 captures high-frequency edge textures; Layer 3 encodes macroscopic object part semantics.

Spatial Alignment & Concatenation

- Bilinear / Adaptive Upsampling: Layer 3 tensor (B, 1024, 14, 14) is upsampled to match Layer 2 grid (B, 1024, 28, 28).
- Channel-wise Concatenation: Produces unified feature tensor phi(x) in R^{B x 1536 x 28 x 28}.
- Spatial Patch Tokenization: Unfolded into 784 localized patch tokens P = {p_i in R^{1536} | i = 1, ..., 784}.
- Zero Training Backpropagation: Pretrained weights frozen; deterministic, reproducible inference encoder.

4. Dimensionality Reduction: Johnson-Lindenstrauss Lemma

Mathematical Formulation & Metric Preservation

- JL Lemma: For any eps in (0, 1) and n vectors in R^D, there exists linear map Psi: R^D -> R^d (d = O(eps^{-2} log n)) such that:
- (1 - eps) ||u - v||_2^2 <= ||Psi u - Psi v||_2^2 <= (1 + eps) ||u - v||_2^2 for all u, v in P.
- Sparse Random Matrix: Entries Psi_{i,j} sampled from Achlioptas distribution: +sqrt(3) with p=1/6, 0 with p=2/3, -sqrt(3) with p=1/6.
- Compression Ratio: Reduces patch vector dimension from 1,536 down to 128 (12x compression) while bounding distance distortion.

Computational Complexity Impact

- Pairwise Distance FLOPs: Scaled from 1,536 multiplications per pair down to 128 FLOPs during coreset candidate evaluation.
- Random Projection Speed: Implemented via dense GPU matrix multiplication Psi @ X^T in <1.2 seconds for 343,000 vectors.
- Zero Accuracy Degradation: Preserves the relative ranking of k-nearest neighbors with >99.4% rank fidelity.

5. Algorithmic Innovation: Minimax Coreset Subsampling

Minimax Facility Location Formulation

- Optimization Objective: Find subset S* subset M with |S*| = k that minimizes the maximum coverage radius:
- S* = argmin_{S subset M, |S|=k} max_{x in M} min_{s in S} ||Psi x - Psi s||_2.
- NP-Hard Problem: Solved via greedy 2-approximation iterative farthest-point sampling.
- Initialization: s_1 chosen uniformly at random from nominal patch collection M.

Iterative Greedy Algorithm & Compression Ratio

- Step t+1: Select s_{t+1} = argmax_{x in M} min_{s in S_t} ||Psi x - Psi s||_2 until |S| = 0.10 * |M|.
- Memory Bank Reduction: Compresses 343,000+ normal patches down to 34,304 prototypes (90% reduction).
- RAM Footprint: Decreases uncompressed memory bank size from 2.1 GB to 210 MB.
- Empirical Validation: Zero loss in detection accuracy (Juice Bottle Image ROC-AUC remains identical at 0.896).

6. Parametric Statistical Modeling: PaDiM Gaussian Estimator

Coordinate-Specific Gaussian Formulation

- Local Parametric Prior: At each spatial patch location (i, j) in [1, 28] x [1, 28], fit a multivariate Gaussian N(mu_{i,j}, Sigma_{i,j}).
- Sample Mean Vector: hat{mu}_{i,j} = (1 / N) * sum_{k=1}^N p_{i,j}^{(k)} in R^{1536}.
- Sample Covariance Matrix: hat{Sigma}_{i,j} = (1 / (N - 1)) * sum_{k=1}^N (p_{i,j}^{(k)} - hat{mu})(p_{i,j}^{(k)} - hat{mu})^T.
- Covariance Regularization: Adds Tikhonov stabilizer (Sigma + eps * I) with eps = 0.01 to ensure non-singular invertibility.

Localized Mahalanobis Distance Metric

- Inference Score: d_M(p_{i,j}) = sqrt((p_{i,j} - hat{mu}_{i,j})^T (hat{Sigma}_{i,j} + eps * I)^{-1} (p_{i,j} - hat{mu}_{i,j})).
- Coordinate Anchoring: Explicitly tests whether a patch deviates from nominal statistics at that exact physical coordinate.
- Comparative Finding: While computationally fast, PaDiM exhibits high sensitivity to mechanical jitter and conveyor vibration.
- PatchCore Dominance: PatchCore's permutation-invariant memory bank achieves higher generalizability (+0.084 ROC-AUC).

7. Inference Engine: Chunked GPU TorchNN Search

Euclidean Distance Decomposition

- Distance Identity: ||u - v||^2 = ||u||^2 + ||v||^2 - 2 * u^T v.
- Precomputation: Precompute nominal norms ||m_j||^2 in R^{34304}; query norm ||p_i||^2 computed on-the-fly.
- Matrix Multiplication: Core computational cost reduced to single batched GEMM: -2 * P @ M^T via NVIDIA Tensor Cores.
- Chunked Query Streaming: Queries processed in chunks of 256 patches to bound peak GPU VRAM allocation under 350 MB.

High-Throughput Latency Profile

- GPU Search Time: 784 test patches queried against 34,304 prototype vectors in 18.4 ms on NVIDIA RTX 5070.
- Full Pipeline Latency: Preprocessing (8ms) + WideResNet (38ms) + TorchNN (18ms) + Heatmap (15ms) = ~85 ms (11.8 FPS).
- CPU Baseline Comparison: Intel Core Ultra 7 CPU executes same pipeline in ~340 ms (4x GPU speedup).
- Industrial Capacity: Sub-100ms round-trip comfortably supports high-speed packaging conveyors at 700 parts/min.

8. Spatial Anomaly Map Generation, Upsampling & Smoothing

Pixel-Level Heatmap Synthesis

- Patch Distance Map: Array of 784 patch scores reshaped to 2D spatial anomaly grid D in R^{28 x 28}.
- Bi-Cubic Spline Upsampling: Interpolates grid D up to native image resolution D_up in R^{256 x 256}.
- 2D Gaussian Kernel Convolution: A(x, y) = (G_sigma * D_up)(x, y) where G_sigma is Gaussian kernel with sigma = 4.0.
- Noise Suppression: Eliminates spurious single-pixel sensor noise while preserving macroscopic defect contours.

Image-Level Anomaly Score Aggregation

- Top-k Softmax Weighting: Rather than taking noisy argmax, aggregate top-k patches: s(X) = (1/k) * sum_{i=1}^k d_{(i)}.
- Receptive Field Normalization: Aggregation parameter k=9 corresponds to spatial area of ~1 receptive field.
- Z-Score Standardization: z = (s(X) - mu_val) / sigma_val for cross-category decision comparability.
- Localization AUROC: Achieved 0.872 pixel-level localization AUROC on the MVTec LOCO AD Juice Bottle benchmark.

9. Statistical Decision Boundary Calibration

Neyman-Pearson Risk Formulation

- Zero Defect Exposure: Detection threshold tau calibrated exclusively on defect-free validation split D_val.
- Statistical Threshold Model: tau_k = mu_val + k * sigma_val, where mu_val, sigma_val are sample moments of nominal scores.
- Multi-Point Operating Envelope: Formulated 3 distinct industrial operating points to align with risk appetite.
- ROC & PR Integration: Threshold sweeps evaluate full Precision-Recall and False Positive Rate curves.

Quantitative Operational Operating Points

- 1. High Sensitivity (k = 1.0, tau = 0.285): Recall = 96.2%, Precision = 78.4%. Zero defect escapes for medical/aerospace lines.
- 2. Optimal F1 Point (k = 2.1, tau = 0.312): Recall = 88.0%, Precision = 87.5%, F1 = 87.7%. Balanced automated production.
- 3. Conservative Point (k = 3.0, tau = 0.354): Recall = 46.6%, Precision = 99.1%. False Alarm Rate < 0.9% for screening.
- Dynamic Recalibration: Operating thresholds adjusted via runtime API payload without retraining model weights.

10. Multi-Category Empirical Benchmark Results

MVTec LOCO AD 5-Category Performance Matrix

- Juice Bottle (Primary): Image ROC-AUC: 0.896 | PR-AUC: 0.941 | Pixel AUROC: 0.872 | Optimal F1: 0.877.
- Hardware Screw Bag: Image ROC-AUC: 0.878 | PR-AUC: 0.912 | Optimal F1: 0.854.
- Pushpins Box: Image ROC-AUC: 0.865 | PR-AUC: 0.895 | Optimal F1: 0.841.
- Splicing Connectors: Image ROC-AUC: 0.861 | PR-AUC: 0.884 | Optimal F1: 0.838.
- Breakfast Box: Image ROC-AUC: 0.842 | PR-AUC: 0.880 | Optimal F1: 0.825.

Cross-Category Generalization Insights

- Mean Image ROC-AUC: 0.868 across diverse domains (liquid bottling, rigid plastic, loose metal fasteners).
- Robust Localization: Pixel AUROC > 0.85 across all categories with precise bounding box enclosure.
- Dominant Failure Modes: Heavily clustered in logical categories involving complex spatial counting constraints.

11. Theoretical Analysis: The 'Logical Anomaly Gap'

Mathematical Cause of the Detection Disparity

- Empirical Gap: Structural Recall = 53.0% vs Logical Recall = 21.8% at conservative threshold tau = mu + 3*sigma (31.2% gap).
- Permutation Invariance Proof: The nearest-neighbor distance operator d(p_i) = min_{m in M} ||p_i - m||_2 is invariant to spatial coordinate (x_i, y_i).
- Missing Part Paradox: When a label is missing, the bare bottle glass patch matches normal glass elsewhere in the memory bank.
- Result: Local patch anomaly score is ~0.0 despite the product violating a global logical assembly rule.

Architectural Solutions to Bridge the Gap

- 1. Spatial Coordinate Conditioning: Appending normalized (x/W, y/H) coordinates to patch embeddings p_tilde = [p_i; gamma * c_i].
- 2. Dual-Threshold Decision Engine: Calibrating separate thresholds for high-frequency structural vs low-frequency logical defects.
- 3. Relational Graph Neural Networks (Future): Formulating spatial relation edges e_{i,j} to model global component topology.

12. Ablation Studies & Comparative Benchmarking

Backbone & Feature Extraction Ablation

- ResNet-18 (Baseline): Image ROC-AUC = 0.812 | Uncompressed RAM = 1.1 GB | Inference = 42 ms.
- WideResNet-50-2 (Layer 2 Only): Image ROC-AUC = 0.841 | Captures fine texture, misses high-level structures.
- WideResNet-50-2 (Layer 3 Only): Image ROC-AUC = 0.856 | Captures global layout, misses micro-scratches.
- WideResNet-50-2 (Layer 2 + Layer 3 Concatenated): Image ROC-AUC = 0.896 | Optimal multi-scale representation.

Coreset Subsampling Ratio Ablation

- 100% Uncompressed (343k patches): ROC-AUC = 0.896 | RAM = 2.1 GB | Latency = 2,400 ms (Unviable in production).
- 10% Greedy Coreset (34k patches): ROC-AUC = 0.896 | RAM = 210 MB | Latency = 85 ms (Ideal Operating Point).
- 1% Greedy Coreset (3.4k patches): ROC-AUC = 0.852 | RAM = 21 MB | Latency = 14 ms (Severe accuracy drop).
- 10% Uniform Random Subsampling: ROC-AUC = 0.881 | Subsampling without minimax facility location loses defect boundary prototypes.

13. Defect Localization & Bounding Box Regression

Pixel-Level Segmentation Algorithm

- Binarization: Apply localized segmentation threshold: M(x, y) = 1 if A(x, y) >= tau_{loc} else 0.
- Morphological Filtering: Sequential morphological opening (kernel 3x3) and closing (kernel 5x5) to eliminate isolated sensor noise.
- Connected Component Labeling (CCL): Two-pass CCL identifies distinct anomaly blobs and computes spatial component statistics.
- Component Saliency: Filter out components with pixel area < 15 pixels to prevent false positive micro-alarms.

Automated Bounding Box Extraction

- Coordinate Regression: Calculate bounding box coordinates enclosing the peak anomaly component: (x_min, y_min, w, h).
- Confidence Tagging: Component confidence c = max_{(x,y) in C_k} A(x, y) overlaid on diagnostic UI.
- Localization Accuracy: Achieves >88% Intersection-over-Union (IoU) with ground truth structural defect masks.
- Interactive Inspection: Coordinates serialized to JSON API response for real-time frontend canvas rendering.

14. Full-Stack Concurrency & Software Engineering

Thread-Safety & Race Condition Prevention

- The Concurrency Hazard: PyTorch forward hooks register intermediate activations globally; concurrent API requests overwrite hook state.
- Mutex Synchronization: Implemented threading.Lock() around model forward pass, ensuring thread-safe execution in FastAPI async workers.
- Defensive Memory Management: Explicit torch.cuda.empty_cache() and garbage collection prevent VRAM fragmentation.
- Windows Smart App Control Compliance: Avoided unverified native DLL binaries by implementing pure Python cryptographic routines.

Asynchronous Backend & Persistent Storage

- FastAPI Asynchronous Microservice: Non-blocking I/O event loop handling /api/inspect, /api/auth, /api/history.
- Cryptographic Security: SQLite database (inspector.db) with PBKDF2-HMAC-SHA256 password hashing and salt.
- Audit Trail Logging: Persistent logging of inspection timestamps, category, z-scores, decision flags, and execution latency.
- Zero External Cloud Dependencies: 100% on-premises offline capability ensuring strict enterprise data security.

15. Explainable AI: 4-View Visual Diagnostic HMI

4-View Visual Diagnostic Decomposition

- View 1: Original High-Resolution RGB Ingestion Frame (256x256 native display).
- View 2: Pinpointed Bounding Box tagging the highest-confidence defect region with coordinate labels.
- View 3: Jet Thermal Colormap illustrating continuous pixel-level anomaly intensity (0.0 to 1.0).
- View 4: Alpha-Blended Overlay merging heatmap with physical part (alpha=0.45) for instant human verification.

End-to-End Latency Budget Breakdown

- 1. Image Ingestion & Tensor Normalization: 8.2 ms.
- 2. WideResNet-50-2 Multi-Scale Feature Extraction: 38.6 ms.
- 3. Chunked GPU TorchNN Nearest-Neighbor Search: 18.4 ms.
- 4. Spatial Interpolation & 2D Gaussian Smoothing: 15.1 ms.
- 5. Total End-to-End Pipeline Latency: 80.3 ms (~12.4 FPS on NVIDIA RTX 5070 GPU).

16. Hardware Scaling & Profiling Analysis

Hardware Compute Benchmarks

- Intel Core Ultra 7 258V (CPU Only): Latency = 342 ms | Throughput = 2.9 FPS | Memory = 420 MB RAM.
- NVIDIA RTX 5070 Laptop GPU (Full FP32): Latency = 125 ms | Throughput = 8.0 FPS | Memory = 650 MB VRAM.
- NVIDIA RTX 5070 Laptop GPU (Chunked TorchNN): Latency = 85 ms | Throughput = 11.8 FPS | Memory = 380 MB VRAM.
- Efficiency Gain: Chunked tensor execution reduces peak VRAM allocation by 41.5% with 1.47x speedup.

Industrial Deployment Feasibility

- Production Line Integration: Easily handles standard packaging conveyor speeds (300 - 700 parts per minute).
- Edge Device Compatibility: Modest 380 MB VRAM footprint enables deployment on compact edge AI devices (NVIDIA Jetson Orin).
- Energy Efficiency: Average GPU power draw under inference load: ~38 Watts.

17. Conclusion & Future Technical Roadmap

Key Technical Contributions Delivered

- 1. Multi-scale feature representation tapping WideResNet Layers 2 & 3 (1,536-D patch tokens).
- 2. 90% memory bank compression via Greedy Minimax Coreset Subsampling with zero loss in ROC-AUC (0.896).
- 3. GPU-accelerated chunked TorchNN engine achieving real-time inference latency of ~85 ms (11.8 FPS).
- 4. Uncovered, formalized, and empirically quantified the 'Logical Anomaly Gap' in nearest-neighbor visual search.
- 5. Production-grade full-stack web application with thread-safe execution locks and 4-panel visual diagnostic explainability.

Future Architectural Roadmap

- Relational Graph Neural Networks (GNNs): Explicitly modeling spatial constraint edges to eliminate the Logical Anomaly Gap.
- Vision Transformer (ViT) Backbones: Leveraging global self-attention to encode long-range inter-component dependencies.
- Multi-Modal Sensor Integration: Transferring coreset distance scoring to high-frequency industrial vibration and thermal telemetry.